

Week 9 Natural Language Processing

Day 1: Introduction to NLP & Text Preprocessing

Theoretical Learning

- What is NLP and its importance in AI.
- Key steps in **text preprocessing**:
 - Noise removal (punctuation, stopwords, HTML tags).
 - Lexicon normalization (lemmatization vs. stemming).
 - Object standardization (lowercasing, expanding contractions).
- Why preprocessing is essential for better model performance.

Study Material

- [Introduction to NLP – GeeksforGeeks](#)
- [Text Preprocessing in NLP – Analytics Vidhya](#)
- [Lemmatization vs. Stemming – NLTK Docs](#)
- [Noise Removal & Text Cleaning – Towards Data Science](#)

Practical Task

- Load a text dataset (e.g., IMDb movie reviews or news dataset).
 - Perform preprocessing:
 - Tokenization.
 - Stopword removal.
 - Lemmatization & stemming.
 - Compare outputs of lemmatization vs. stemming with examples.
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Day 2: Feature Engineering on Text Data

Theoretical Learning

- Bag of Words (BoW).
- N-Grams (bigrams, trigrams).
- Statistical features (word counts, sentence length).

- **TF-IDF** and why it's better than raw frequency.
- Readability features (Flesch reading ease, etc.).

Study Material

- [Bag of Words Model – Scikit-Learn Docs](#)
- [TF-IDF Explained – GeeksforGeeks](#)
- [N-Grams in NLP – Towards Data Science](#)
- [Text Feature Extraction with Python – MachineLearningMastery](#)

Practical Task

- Convert preprocessed dataset into BoW and TF-IDF features.
 - Visualize top 10 most frequent words.
 - Train a **Naive Bayes classifier** on BoW and TF-IDF representations.
 - Compare accuracy, precision, and recall.
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Day 3: Syntactical & Semantic Analysis

Theoretical Learning

- Part-of-Speech (POS) tagging.
- Syntactic parsing (dependency grammar).
- Named Entity Recognition (NER).
- Phrase detection & entity parsing.
- Semantic similarity measures:
 - Levenshtein distance.
 - Phonetic matching.
 - Flexible string matching.

Study Material

- [Part of Speech Tagging – NLTK Guide](#)
- [Dependency Parsing – spaCy Docs](#)
- [Named Entity Recognition \(NER\) – spaCy Docs](#)
- [Levenshtein Distance – GeeksforGeeks](#)

- [Coreference Resolution – Hugging Face](#)

Practical Task

- Use **spaCy** or **NLTK** for:
 - POS tagging and dependency parsing.
 - Named Entity Recognition (NER).
 - Implement Levenshtein distance for two sentences and test similarity.
 - Show examples of coreference resolution (using spaCy/NeuralCoref).
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Day 4: Word Embeddings & Topic Modeling

Theoretical Learning

- Limitations of BoW/TF-IDF (no context).
- Word Embeddings: Word2Vec, GloVe, FastText.
- Transfer learning in NLP (pre-trained embeddings).
- Topic Modeling with Latent Dirichlet Allocation (LDA).

Study Material

- [Word Embeddings – Stanford CS224N Lecture Notes](#)
- [Word2Vec Tutorial – TensorFlow](#)
- [Gensim Word2Vec Implementation](#)
- [Topic Modeling with LDA – MachineLearningMastery](#)
- [Word Embeddings vs. BoW – Medium Article](#)

Practical Task

- Train a Word2Vec model using **gensim** on your dataset (or use pre-trained embeddings).
- Visualize word vectors in 2D (e.g., "king – man + woman $\approx$ queen").
- Implement topic modeling (LDA) on a dataset and extract top topics.
- Compare BoW vs. Word2Vec performance on text classification.

Day 5: Deep Learning for NLP (RNNs & LSTMs)

Theoretical Learning

- Introduction to sequence models (RNN, LSTM).
- Why RNNs struggle with long sequences (vanishing gradient).
- How LSTMs solve this problem.
- Applications: text classification, sentiment analysis, machine translation.

Study Material

- [Understanding RNNs & LSTMs – Colah’s Blog](#)
- [Recurrent Neural Networks – DeepLearning.ai](#)
- [Sentiment Analysis with LSTM – TensorFlow Tutorial](#)
- [PyTorch RNN/LSTM Example](#)
- [Transfer Learning in NLP – Hugging Face Transformers](#)

Practical Task

- Build a **sentiment analysis model** using RNN/LSTM in Keras/PyTorch.
- Train on IMDb reviews dataset with Word2Vec embeddings.
- Evaluate using accuracy, F1-score, and confusion matrix.
- Save the trained model for reuse.